Bending free energy with renormalized area

OR

How Safran does it

OR

What is Σ? ← "Zeta"

For small H, we can expand the elastic free energy like so:

$$f_{el} = f_0 + f_1 H + f_2 H^2 + O(H^3)$$

even if $c_0 \neq 0$. However, f_0, f_1, f_2 could be dependent on Σ. Let's assume they are and see what we get:

$\Delta\Sigma = \Sigma - \Sigma_0$!

$f_0 = f_{00} + f_{01}\Delta\Sigma + \frac{1}{2} f_{02}\Delta\Sigma^2 + O(\Delta\Sigma)^3$

$f_1 = f_{10} + f_{11}\Delta\Sigma + O(\Delta\Sigma^2)$

$f_2 = f_{20} + O(\Delta\Sigma)$

original term	0^{th} order	1^{st} order	2^{nd} order
f_0	f_{00}	$f_{01}(\Delta\Sigma)$	$\frac{1}{2} f_{02}(\Delta\Sigma)^2$
$f_1 H$	—	$f_{10} H$	$f_{11}\Delta\Sigma H$
$f_2 H^2$	—	—	$f_{20} H^2$

Total, we get:

$$F_{el} = f_{00} + \cancel{f_{01}}^{\,0}\Delta\Sigma + \frac{1}{2} f_{02}\Delta\Sigma^2 + f_{10} H + f_{11}\Delta\Sigma H + f_{20} H^2$$

Previously:

$$f_{ee} = f_{00} + \frac{1}{2} f_{02} \Delta\Sigma^2 + f_{10} H + f_{11} \underbrace{\Delta\Sigma H}_{} + f_{20} H^2$$

"coupling term" between H and Σ. The minimizing $\Delta\Sigma$ will depend on H.

$\boxed{\text{What Safran does Next}}$

Let's find $\Delta\Sigma^* = \Sigma^* - \Sigma_0$, where Σ^* is the minimizing area for finite curvature:

$$\left.\frac{\partial f_{ee}}{\partial \Delta\Sigma}\right|_{\Delta\Sigma^*} = f_{02}\Delta\Sigma^* + f_{11} H = 0$$

$$\Delta\Sigma^* = -\frac{f_{11}}{f_{02}} H$$

Q1: ← what's a sanity check for this expression?

Assuming the membrane can adjust Σ as a function of H, the curvature free energy is:

$$f = f_{ee}(\Delta\Sigma^*) = f_{00} + \frac{1}{2} f_{02}\left(\frac{-f_{11} H}{f_{02}}\right)^2 + f_{20} H^2 - \frac{f_{11}^2}{f_{02}} H^2 + f_{10} H$$

$$f = f_{00} + f_{10} H + \underbrace{\left(f_{20} - \frac{f_{11}^2}{2f_{02}}\right)}_{} H^2$$

Now at this point, Safran redefines stuff

g_0 g_1 g_2

$$f = g_0 + g_1 H + g_2 H^2$$

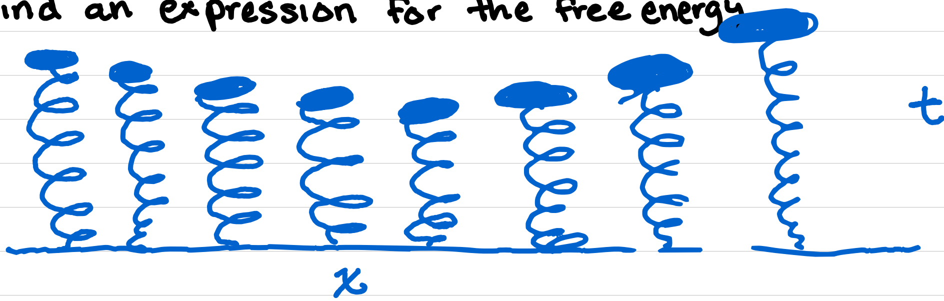
$$g_2 \rightarrow 2k_c$$

$$g_1/2g_2 \rightarrow c_0$$

But suppose instead of minimizing Σ, we just let it vary.

Example Application:

Lipids adhering to a surface:
Find an expression for the free energy



Each lipid has an area Σ_i, thickness t_i, curvature H_i and we can sum over the lipids to get the total F.e

$$F_{el} = \sum_i f_{el,i} = \sum_i \frac{f_{02}}{2}(\Delta\Sigma_i)^2 + f_{1,0} H_i + f_{11} H_i \Delta\Sigma_i + f_{20} H_i^2$$

but an integral version will be more tractable:

$$\sum_i f_{el,i} \rightarrow \int dxdy\, f_{el}(x,y) / \Sigma(x,y)$$

That is: (Σ, $\Delta\Sigma$, H are all functions of x and y)

$$F_{el} = \int dxdy \left\{ f_{02}\frac{(\Delta\Sigma)^2}{\Sigma} + f_{10}\frac{H}{\Sigma} + f_{11} H \frac{\Delta\Sigma}{\Sigma} + f_{20}\frac{H^2}{\Sigma} \right\}$$

To deal with all these Σs, use volume conservation:

$$t_0\Sigma_0 = t\Sigma \rightarrow t_0\Sigma_0 = [t_0 + \overbrace{(t-t_0)}^{\Delta t}]\Sigma = (t_0 + \Delta t)\Sigma$$

$$\Sigma_0 = (1 + \Delta t / t_0)\Sigma$$

Identities and Approximations

Term	Equivalent term (to 2^{nd} order)	
Σ_0	$(1 + \Delta t / t_0)\Sigma$	volume conservation
$1/\Sigma$	$(1 + \Delta t / t_0)/\Sigma_0$	rearrangement
$\Delta\Sigma$	$-(\Delta t / t_c)\Sigma$	$\Delta\Sigma = \Sigma - \Sigma_0$
$\dfrac{(\Delta\Sigma)^2}{\Sigma}$	$(\Delta t / t_0)^2 \Sigma_0$	drop 3rd order terms like $(\Delta t^2 \Delta\Sigma)$
H/Σ	$\frac{1}{2}(\nabla^2 t)(1 + \Delta t / t_0)/\Sigma_0$	Monge gauge
$\dfrac{H\Delta\Sigma}{\Sigma}$	$-\frac{1}{2}(\nabla^2 t)(\Delta t / t_0)$	
$\dfrac{H^2}{\Sigma}$	$\dfrac{1}{4}\dfrac{(\nabla^2 t)^2}{\Sigma_0}$	Drop 3rd order terms like $(\nabla^2 t)^2 \Delta t$

Exercise: Verify each identity

Using the substitutions/identities in the previous table,

$$F = \int dx\,dy \left\{ f_{0,2} \frac{(\Delta\Sigma)^2}{\Sigma} + f_{1,0} \frac{H}{\Sigma} + f_{1,1} \frac{H \Delta\Sigma}{\Sigma} + f_{2,0} \frac{H^2}{\Sigma} \right\}$$

becomes

$$F = \int dx\,dy \left\{ f_{02} \frac{\Sigma_0 (\Delta t)^2}{t_0^2} + \frac{f_{1,0}}{2\Sigma_0} \nabla^2 t \left(1 + \frac{\Delta t}{t_0}\right) - f_{1,1} \frac{\nabla^2 t}{2} \frac{\Delta t}{t_0} + \frac{f_{2,0}}{4} \frac{(\nabla^2 t)^2}{\Sigma_0} \right\}$$

or, rearranging,

$$F = \int dx\,dy \left\{ f_{02} \frac{\Sigma_0}{t_0^2} (\Delta t)^2 + \frac{f_{1,0}}{2\Sigma_0} \nabla^2 t + \frac{1}{2\Sigma_0 t_0} (f_{10} - \Sigma_0 f_{11}) \nabla^2 t \, \Delta t + \frac{f_{20}}{4} \frac{(\nabla^2 t)^2}{\Sigma_0} \right\}$$

So we can give all these constants more traditional names:

$$f_{0,2}\Sigma_0 \to \frac{k_A}{2} \qquad \frac{f_{1,0}}{\Sigma_0} \to 2k_c c_0 \qquad \frac{f_{1,1}}{\Sigma_0} \to 2k_c c_0' \qquad \frac{f_{2,0}}{\Sigma_0} \to k_c$$

area derivative of c_0

so,

$$F = \int dx\,dy \left\{ \frac{k_A}{2} \left(\frac{\Delta t}{t_0}\right)^2 + k_c c_0 \nabla^2 t + k_c (c_0 - c_0' \Sigma_0) \nabla^2 t \left(\frac{\Delta t}{t_0}\right) + \frac{k_c}{4} (\nabla^2 t)^2 \right\}$$

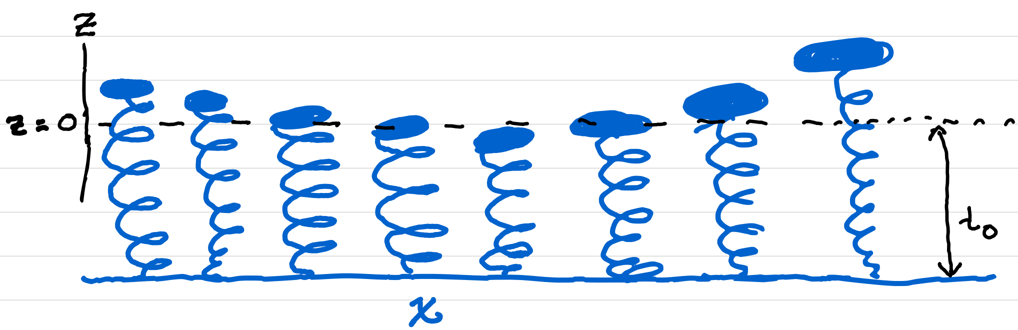
define $\xi = c_0 - c_0' \Sigma$ and that's where ξ comes from!

So now we have an expression just in terms of t!

Well, actually, it's in terms of $\nabla^2 t$ and Δt.

But $\nabla^2 t = \nabla^2(\Delta t)$! So we can put the whole expression in terms of Δt. But $\nabla^2(\Delta t)$ looks terrible. Let's give Δt a new name, z, and then:

$$F = \int dx\,dy \left(\frac{k_A}{2}\frac{z^2}{t_0^2} + k_c c_0 \nabla^2 z + k_c \xi \frac{z}{t_0}\nabla^2 z + \frac{k_c}{4}(\nabla^2 z)^2 \right)$$



Final thought... why $\xi = c_0 - c_0' \Sigma_0$ instead of just c_0? What will happen if ξ is negative?